Dr. Harender S. Dhattarwal

Postdoctoral Associate
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Summary

Expertise in the bulk and interfacial molecular dynamic simulations of simple and complex liquids. Strong background in classical, quantum, and machine learning potential-based molecular dynamic simulations, including experience in quantitative data management. Skilled in high performance computing.

Education

Ph.D. in Computational Chemistry — *July 2022*
Indian Institute of Technology Delhi — *Delhi, India*
PI: Prof. Hemant K. Kashyap (IIT Delhi, India), Co PI: Prof. Jer-Lai Kuo (IAMS, Taiwan)

Master of Science in Chemistry — *July 2017*
Indian Institute of Technology Guwahati — *Guwahati, India*

Bachlors of Science in Chemistry — *July 2015*
University of Delhi — *Delhi, India*

Research Work

- Neural network potentials for molecular dynamics simulations with long-range interactions
- Conduction mechanism in solid-state superionic conductors
- Mechanism of solid electrolyte interphase (SEI) formation near a metal anode
- Mechanistic insights on electrolyte intercalation-deintercalation in nanoscale confinements
- Structure and dynamics of ionic liquids, water-in-salt, and hybrid electrolytes in bulk and near interface
- Wetting-dewetting behavior of carbon surfaces by ionic liquids and deep eutectic solvents (DES)

Experience

Postdoctoral Associate — *July 2022 - Present*
Department of Chemistry and Chemical Biology, Rutgers University — *New Brunswick, NJ, USA*

Summer Intern — *May 2019 - June 2019*
Institute of Atomic and Molecular Sciences, Academia Sinica — *Taipei, Taiwan*

Winter Intern — *Oct. 2018 - Nov. 2018*
Institute of Atomic and Molecular Sciences, Academia Sinica — *Taipei, Taiwan*

Skills

Programming	Python, Fortran, C++, Bash scripting
Scientific	CP2K, GROMACS, LAMMPS, GAUSSIAN, Gauss View
Graphics	VMD, Xmgrace, Avogadro, Gnuplot, Matlab, Vector Graphics

Publications

1. **Harender S. Dhattarwal** and Hemant K. Kashyap, Microstructures of Choline Amino Acid based Biocompatible Ionic Liquids, *Chem. Record* **2023**,e202200295. [Invited Review]
2. **Harender S. Dhattarwal**, Ang Gao, and Richard C. Remsing, Dielectric Saturation in Water from a Long Range Machine Learning Model. *arXiv* **2023**.
3. **Harender S. Dhattarwal** and Hemant K. Kashyap, Heterogeneity and Nanostructure of Superconcentrated LiTFSI–EmimTFSI Hybrid Aqueous Electrolytes: Beyond the 21 m Limit of Water-in-Salt Electrolyte. *J. Phys. Chem. B* **2022** 126 (28), 5291.
4. **Harender S. Dhattarwal**, Jer-Lai Kuo, and Hemant K. Kashyap, Mechanistic Insight on the Stability of Ether and Fluorinated Ether Solvent-Based Lithium Bis (fluoromethanesulfonyl) Electrolytes near Li Metal Surface. *J. Phys. Chem. C* **2022** 126 (20), 8953.
5. **Harender S. Dhattarwal** and Hemant K. Kashyap, Unique and Generic Structural Features of Cholinium Amino Acid Biocompatible Ionic Liquids. *Phys. Chem. Chem. Phys.* **2021** 23, 10662.
6. **Harender S. Dhattarwal**, Richard C. Remsing, and Hemant K. Kashyap, Intercalation–deintercalation of water-in-salt electrolytes in nanoscale hydrophobic confinement. *Nanoscale* **2021** 13, 4195.
7. **Harender S. Dhattarwal**, Yun-Wen Chen, Jer-Lai Kuo and Hemant K. Kashyap, Mechanistic Insight on the Formation of Solid Electrolyte Interphase (SEI) by Acetonitrile-Based Superconcentrated [Li][TFSI] Electrolyte Near Li Metal. *J. Phys. Chem. C* **2020** 124, 27495.
8. **Harender S. Dhattarwal**, Richard C. Remsing and Hemant K. Kashyap, How Flexibility of Nanoscale Solvophobic Confining Material Promotes Capillary Evaporation of Ionic Liquids, *J. Phys. Chem. C* **2020** 124, 4899.
9. **Harender S. Dhattarwal** and Hemant K. Kashyap, Molecular Dynamics Investigation of Wetting-Dewetting Behavior of Model Carbon Material by 1-Butyl-3-methylimidazolium Acetate Ionic Liquid Nanodroplet, *J. Chem. Phys.* **2019** 151, 244705.
10. Navneet Singh, Mrityunjay K. Jha, **Harender S. Dhattarwal** and Hemant K. Kashyap, How NaFTA Salt Affects the Structural Landscape and Transport Properties of $Pyrr_{1,3}$FTA Ionic Liquid. *J. Chem. Phys.* **2023** 158, 104502.
11. Akshay Malik, **Harender S. Dhattarwal** and Hemant K. Kashyap, An Overview of Structure and Dynamics Associated with Hydrophobic Deep Eutectic Solvents and Their Applications in Extraction Processes. *ChemPhysChem* **2022** 23 (18), e202200239.
12. Akshay Malik, **Harender S. Dhattarwal** and Hemant K. Kashyap, Distinct Solvation Structures of CO_2 and SO_2 in Reline and Ethaline Deep Eutectic Solvents Revealed by AIMD Simulations. *J. Phys. Chem. B* **2021** 125, 1852.
13. Aditya Gupta, **Harender S. Dhattarwal**, and Hemant K. Kashyap, Structure of Cholinium Glycinate Biocompatible Ionic Liquid at Graphite Electrode Interface. *J. Chem. Phys.* **2021** 54, 184702.
14. Vikas Khatri, **Harender S. Dhattarwal**, Hemant K. Kashyap, and Gurmeet Singh, First-principles based Theoretical Investigation of Impact of Polyolefin Structure on Photo-oxidation Behaviour. *J. Comp. Chem.* **2021** 42, 1710.
15. Akshay Malik, **Harender S. Dhattarwal** and Hemant K. Kashyap, Molecular Dynamics Investigation of Wetting-Dewetting Behavior of Reline DES Nanodroplet at Model Carbon Material. *J. Chem. Phys.* **2020** 153, 164704.
16. Shobha Sharma, **Harender S. Dhattarwal** and Hemant K. Kashyap, Molecular Dynamics Investigation of Electrostatic Properties of Pyrrolidinium Cation Based Ionic Liquids Near Electrified Carbon Electrodes, *J. Mol. Liq.* **2019** 291, 111269.
17. Akshay Malik, **Harender S. Dhattarwal** and Hemant K. Kashyap, Applications of Deep Eutectic Solvent in Gas Capture. *Current Developments in Biotechnology and Bioengineering (Elsevier)* **2022**, 49-75. [Invited Book Chapter]

18. **Harender S. Dhattarwal** and Hemant K. Kashyap, Structure and Stability of Modern Electrolytes in Nanoscale Confinements from Molecular Dynamics Perspective, *Synthesis and Applications of Nanomaterials and Nanocomposites (Elsevier)* [Invited Book Chapter] (Accepted)

Conferences

1. Oral: Mechanistic Insight on the Solid Electrolyte Interphase (SEI) formed by a Superconcentrated [Li][TFSI] in Acetonitrile Electrolyte near Lithium Metal, *The 240th Electrochemical Society Meeting* (**ECS 2021**).
2. Poster: Insights on the Initial Solid Electrolyte Interphase (SEI) formation by a Superconcentrated Electrolyte at Li Metal Surface, *Theoretical Chemistry Symposium* (**TCS 2021**), IISER Kolkata, Kolkata.
3. Poster: Wetting and de-wetting Behavior of Flexible Carbon-based Electrode Materials by Imidazolium-based Ionic Liquid, *Advanced Simulation Methods: DFT, MD and Beyond* (**ASM 2019**), IIT Delhi, Delhi.
4. Poster: Effect of Flexibility of Confining Materials on Capillary Evaporation of the Ionic Liquids, *Dynamics at the Interface of Chemistry and Biology* (**DICB 2019**), IISC Bangalore, Bengaluru.
5. Poster: Wetting and de-wetting Behavior of Carbon-based Electrode Materials by Imidazolium-based Ionic Liquid: Role of Solvophilicity and Flexibility, *3rd North-West Meeting on Spectroscopy, Structure and Dynamics* (**SSD 2019**), IIT Roorkee, Roorkee.

Awards

- PCCP Best Poster Presentation Award at Theoretical Chemistry Symposium (TCS 2021) IISER Kolkata, Kolkata.
- Best Poster Presentation Award at Advanced Simulation Methods: DFT, MD and Beyond (ASM 2019), IIT Delhi, Delhi.
- Senior Research Fellowship awarded by UGC-India (July 2019–July 2022).
- Junior Research Fellowship awarded by UGC-India (July 2017–July 2019).